

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0435

Swan 138 kV
July 22, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0435. The Interconnection Customer proposes a 122.2 MW battery energy storage facility interconnecting at the Swan 138 kV substation in the Virginia area. The request was received on February 15, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 231 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,974,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Travis Ab - Werner W 138 KV
Contingency: Mccartney - Yarnen 345 KV
Loading: 141.1% (Rating: 259 MVA)
DFAX: 8.5%
Pre-project loading: 137.1%
2. Facility: Floraday - Villa Del Rey 3010 138 KV
Contingency: North Tower Tap - Mccartney 345 KV
Loading: 137.9% (Rating: 289 MVA)
DFAX: 8.6%
Pre-project loading: 134.3%
3. Facility: Deen Pt - Locks 138 KV L
Contingency: Mccartney - N Tower Tap 345 KV
Loading: 137.0% (Rating: 169 MVA)
DFAX: 7.0%
Pre-project loading: 131.9%
4. Facility: Walthall Substation - Unknown 138 KV
Contingency: N Tower Tap - Haggin 345 KV L
Loading: 132.0% (Rating: 317 MVA)
DFAX: 8.0%
Pre-project loading: 128.9%
5. Facility: N Tower Tap Substation 345 kV/138 kV Transformer
Contingency: Haggin - North Tower Tap 345 KV
Loading: 131.9% (Rating: 467 MVA)
DFAX: 12.2%
Pre-project loading: 128.7%

6. Facility: Floraday - Sonoma 3 138 KV
Contingency: N Tower Tap - McCartney 345 KV
Loading: 130.7% (Rating: 203 MVA)
DFAX: 7.0%
Pre-project loading: 126.5%
7. Facility: Locks - Sonoma 3 138 KV
Contingency: N Tower Tap - McCartney 345 KV line
Loading: 130.2% (Rating: 203 MVA)
DFAX: 7.0%
Pre-project loading: 126.0%
8. Facility: Gosport - Wheelabrator Portsmouth 138 KV
Contingency: Swan - Mickens 138 KV
Loading: 128.7% (Rating: 371 MVA)
DFAX: 50.8%
Pre-project loading: 112.0%
9. Facility: Unknown - North Tower Tap 138 KV
Contingency: North Tower Tap Substation - Haggin 345 KV
Loading: 126.3% (Rating: 344 MVA)
DFAX: 8.0%
Pre-project loading: 123.4%
10. Facility: Hewitt Road - Acec Lewiston 138 KV
Contingency: New Denver - Martin Dam Substation 138 KV
Loading: 121.7% (Rating: 338 MVA)
DFAX: 5.7%
Pre-project loading: 119.7%
11. Facility: York Canyon - Hat Rapids 138 KV
Contingency: White Stone - Meaux 138 KV
Loading: 119.0% (Rating: 589 MVA)
DFAX: 11.1%
Pre-project loading: 116.7%
12. Facility: Wheelabrator Portsmouth - Victory Way Substation 138 KV Line
Contingency: Reeves Ave - Wheelabrator Portsmouth 138 KV
Loading: 117.4% (Rating: 234 MVA)
DFAX: 8.9%
Pre-project loading: 112.8%
13. Facility: Danc Lfge Facility - Elko 138 KV
Contingency: Meaux - White Stone 138 KV line
Loading: 117.0% (Rating: 228 MVA)
DFAX: 6.4%
Pre-project loading: 113.6%
14. Facility: Lamping Station - Daleville 138 KV
Contingency: New Denver Substation - Greening 138 KV
Loading: 116.6% (Rating: 556 MVA)
DFAX: 8.8%
Pre-project loading: 114.7%
15. Facility: Hull Street Road 345/138 kV Transformer
Contingency: Mitchell Dam - Skaggs Island Jct 2 345 KV Line
Loading: 116.0% (Rating: 529 MVA)
DFAX: 13.4%
Pre-project loading: 112.9%
16. Facility: New Denver - Daleville 138 KV
Contingency: New Denver - Greening 138 KV Line

Loading: 113.6% (Rating: 479 MVA)

DFAX: 7.0%

Pre-project loading: 111.8%

17. Facility: Grahamsville - Greening 138 KV

Contingency: Greening 345 kV/138 kV Autotransformer

Loading: 111.2% (Rating: 381 MVA)

DFAX: 5.6%

Pre-project loading: 109.4%

18. Facility: Hewitt Rd - Grahamsville 138 KV

Contingency: Greening 345/138 kV XFMR

Loading: 110.1% (Rating: 386 MVA)

DFAX: 6.1%

Pre-project loading: 108.1%

19. Facility: York Canyon - Martin Dam 138 KV

Contingency: Martin Dam - New Denver 138 KV

Loading: 106.4% (Rating: 450 MVA)

DFAX: 5.6%

Pre-project loading: 104.9%

20. Facility: Riva Rd - Shingletown 345 KV L

Contingency: Taber - Glenannan 345 KV line

Loading: 104.4% (Rating: 1,356 MVA)

DFAX: 8.0%

Pre-project loading: 103.7%

21. Facility: Syl Laskin - Elko Substation 138 KV

Contingency: Haggin - Mitchell Dam 345 KV

Loading: 104.0% (Rating: 251 MVA)

DFAX: 7.0%

Pre-project loading: 100.6%

22. Facility: Taber 345-138 kV Auto

Contingency: Taber Substation - Dixleta 345 KV line

Loading: 102.1% (Rating: 1,107 MVA)

DFAX: 5.4%

Pre-project loading: 101.5%

23. Facility: Toano - Lightfoot 138 KV L

Contingency: White Stone - Meaux 138 KV

Loading: 101.7% (Rating: 335 MVA)

DFAX: 9.3%

Pre-project loading: 98.3%

24. Facility: Roberts Sw Sta - Navoo 345 KV

Contingency: City West Diesel Plant - Nauvoo 345 KV line

Loading: 101.4% (Rating: 1,290 MVA)

DFAX: 7.7%

Pre-project loading: 100.7%

25. Facility: Mccartney - N Tower Tap 345 KV

Contingency: Bear - Mecklenburg Power Sta 345 KV Line

Loading: 101.4% (Rating: 1,435 MVA)

DFAX: 27.0%

Pre-project loading: 99.1%

26. Facility: Bear Island - Dunn Siding 345 KV

Contingency: Leavitt - Birchwood 345 KV Line

Loading: 101.3% (Rating: 1,885 MVA)

DFAX: 24.5%
Pre-project loading: 99.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Werner West - Travis Ab Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$11,373,000	5% (\$609,000)
Floraday - Villa Del Rey 3010 138 KV	Rebuild line with higher-capacity conductor	\$11,168,000	5% (\$527,000)
Locks - Deen Pt 138 KV line	Rebuild line with higher-capacity conductor	\$1,092,000	4% (\$47,000)
Walthall - Unknown 138 KV	Rebuild line with higher-capacity conductor	\$19,226,000	4% (\$807,000)
N Tower Tap Substation 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$12,370,000	4% (\$535,000)
Floraday - Sonoma 3 138 KV	Rebuild line with higher-capacity conductor	\$19,552,000	4% (\$833,000)
Sonoma 3 - Locks 138 KV line	Rebuild line with higher-capacity conductor	\$2,994,000	4% (\$128,000)
Wheelabrator Portsmouth - Gosport 138 KV	Rebuild line with higher-capacity conductor	\$3,084,000	19% (\$594,000)
Unknown - North Tower Tap 138 KV	Rebuild line with higher-capacity conductor	\$5,242,000	4% (\$220,000)
Hewitt Rd - Acec Lewiston 138 KV	Rebuild line with higher-capacity conductor	\$34,639,000	2% (\$746,000)
York Canyon - Hat Rapids 138 KV	Rebuild line with higher-capacity conductor	\$35,150,000	2% (\$574,000)
Wheelabrator Portsmouth - Victory Way Substation 138 KV Line	Rebuild line with higher-capacity conductor	\$9,090,000	6% (\$567,000)
Elko - Danc Lfgte Facility 138 KV	Rebuild line with higher-capacity conductor	\$4,928,000	2% (\$85,000)
Daleville - Lamping Sta 138 KV L	Rebuild line with higher-capacity conductor	\$3,267,000	2% (\$55,000)
Hull Street Road 345/138 kV Transformer	Replace transformer with higher-rated unit	\$18,046,000	3% (\$592,000)
New Denver - Daleville 138 KV	Rebuild line with higher-capacity conductor	\$19,141,000	2% (\$291,000)
Grahamsville - Greening 138 KV	Rebuild line with higher-capacity conductor	\$6,494,000	2% (\$112,000)
Grahamsville - Hewitt Road 138 KV Line	Rebuild line with higher-capacity conductor	\$7,257,000	2% (\$126,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
York Canyon - Martin Dam Substation 138 KV Line	Reconductor line	\$9,911,000	1% (\$128,000)
Riva Rd - Shingletown 345 KV L	Reconductor line	\$36,600,000	2% (\$637,000)
Syl Laskin - Elko Substation 138 KV	Reconductor line	\$1,859,000	3% (\$55,000)
Taber 345-138 kV Auto	Replace transformer with higher-rated unit	\$11,229,000	2% (\$226,000)
Toano - Lightfoot 138 KV	Reconductor line	\$1,184,000	2% (\$29,000)
Roberts Sw Sta - Navoo 345 KV	Reconductor line	\$11,421,000	2% (\$242,000)
Mccartney - North Tower Tap 345 KV	Reconductor line	\$33,419,000	4% (\$1,287,000)
Bear Island - Dunn Siding 345 KV	Reconductor line	\$41,013,000	2% (\$922,000)
	Total allocated to project		\$10,974,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.